Abstract Preview

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A novel chunk-for-pooling machine learning model to detect measurable residual disease (MRD) in acute myeloid leukemia (AML) with flow cytometric data

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Introduction: In the treatment of acute myeloid leukemia (AML) patients, detection of measurable residual disease (MRD) is critical in guiding effective treatment and predicting complete remission. Manual analysis, however, of flow cytometry (FCM) data for AML MRD detection is complex, requiring significant expertise and is time consuming. This study proposed a machine learning (ML) classification model that could automated detecting MRD in AML patients. Method: We analyzed FCM datasets for 1,040 bone marrow specimens collected at Roswell Park Comprehensive Cancer Center between 2015 to 2020, acquired on a BD FACSCanto™ II cytometer. We leveraged a chunk-for-pooling framework to train an ML classifier that distinguished MRD-negative and MRD-positive specimens. We applied this approach on the whole dataset (n=1040) and the normal-cellularity sub-dataset (n=261). Result: The model achieved an area under the ROC curve (AUC) of 88.5%. Incorrect predictions were more probable on specimens with low cellularity (20.5%, n=160) than on normal-cellularity specimens (11.5%, n=30). The model on the sub-dataset outperformed the whole dataset (AUC=92.1%). According to the error analysis, 8.9% of mis-classified cases showed residual disease percentage ≥ 1%. Conclusion: We demonstrated here the potential of the chunk pooling model to support automated AML-MRD detection of large data files. The model performance was even better when the targeted dataset was constrained to specimens with normal cellularity. Immunophenotypes captured by chunks can be helpful for explaining the decision made by the classifier and provide a robust result for clinical use.